Chapter11 DSP Algorithm Implementation

- ***Structure Simulation and Verification by MATLAB***
- ***Computation of DFT***

✓ ***Fast Fourier Transformation —— FFT***

✓ ***DFT and IDFT Computation Using MATLAB***

- ***Number Representation***

11.2 Structure simulation Using MATLAB

- **Program:**

  direct2.m, 11_1.m, 11_2.m, 11_3.m,
  11_4.m, 11_5.m, 11_6.m, 11_7.m

11.3 Computation of the DFT

11.3.1 Goertzel's Algorithm

11.3.2 Cooley-Tukey FFT Algorithms

- **FFT**--Fast Fourier Transformation, not a new transformation, but a fast algorithm to calculate the DFT.
- J.W.Cooley and J.W.Tukey are given credit for bringing the FFT to the world in their paper :“An algorithm for the machine calculation of complex Fourier Series”, Mathematics Computation, Vol.19, 1965.
- IEEE electroacoustic journal published an album on fast Fourier transform algorithms in 1967, starting the history of fast transform.
- Magazine in January 1992 and IEEE ICASSP 92 have featured articles or reports on commemorating the release of FFT albums.
- In April 1994, the album "Understanding Fourier Technique" is published with the purpose of popularizing and promoting FFT technology

Development of FFT

- **The traditional fast Fourier transform algorithm is recursive decomposition algorithm, including basis 2, basis 4, basis 8 and split basis FFT.**
- **The radix K-FFT (also known as convolutional FFT algorithm C-FFT) proposed by Stasinisik is a new development of other radix FFT algorithms.**
- **Frigo and Johnson developed the FFTW free library in 2005 to provide a solution for running the FFT algorithm on a variety of hardware platforms.**

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- **DFT and IDFT definition:**

$$X[k]=\sum_{n=0}^{N-1}x[n]W_N^{kn},0\le k\le N-1$$

$$x[n]=\frac{1}{N}\sum_{k=0}^{N-1}X[k]W_N^{-kn},0\le n\le N-1$$

Where, $W_N=e^{-j2\pi/N}$

- **Direct computation of all *N* samples of {*X*[*k*]} requires N^2 complex multiplications and $N(N-1)$ complex additions.**

The properties of twiddle factor W_N:

- **Symmetry:** $(W_N^{nk})^* = W_N^{-nk}$
- **Periodicity:** $W_N^{nk} = W_N^{(n+N)k} = W_N^{n(k+N)}$
- **Reduction:** $W_N^{nk} = W_{mN}^{nmk} \quad W_N^{nk} = W_{N/m}^{nk/m}$

$$W_N^{n(N-k)} = W_N^{(N-n)k} = W_N^{-nk}, W_N^{N/2} = -1, W_N^{(k+N/2)} = -W_N^k$$

Decimation-in-Time FFT

- Given a sequence x[n] whose length is $N=2^k$, k is an integer.
- Divided the sequence into even index sequences and odd index sequences:

$$\begin{aligned} x[2r] &= x_0[r] \\ x[2r+1] &= x_1[r] \end{aligned}, r = 0,1,2,\ldots,\frac{N}{2}-1$$

- Since $x[n]$ is first decimated to form a set of subsequences before the DFT is computed, this computation schemes are called decimation-in-time (DIT) FFT algorithms.

Decimation-in-Time FFT

$$X[k] = DFT\{x[n]\} = \sum_{n=0}^{N-1} x[n]W_N^{nk} = \sum_{r=0}^{N/2-1} x[2r]W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1]W_N^{(2r+1)k}$$

$$= \sum_{r=0}^{N/2-1} x_0[r]W_N^{2rk} + W_N^k \sum_{r=0}^{N/2-1} x_1[r]W_N^{2rk} = \sum_{r=0}^{N/2-1} x_0[r]W_{N/2}^{rk} + W_N^k \sum_{r=0}^{N/2-1} x_1[r]W_{N/2}^{rk}$$

- **We get:** $X[k] = X_0[k] + W_N^k X_1[k], k = 0,1,2,\ldots N/2-1$

Where $X_0[k]$ and $X_1[k]$ is *N/2*-point DFT, so we get only first half *N/2*-point result of X[k].

Decimation-in-Time FFT

- **Based on the periodicity of W_N, we can get:**

$$X_0[\frac{N}{2}+k]=\sum_{r=0}^{N/2-1}x_0[r]W_{N/2}^{r(\frac{N}{2}+k)}=\sum_{r=0}^{N/2-1}x_0[r]W_{N/2}^{rk}=X_0[k]$$

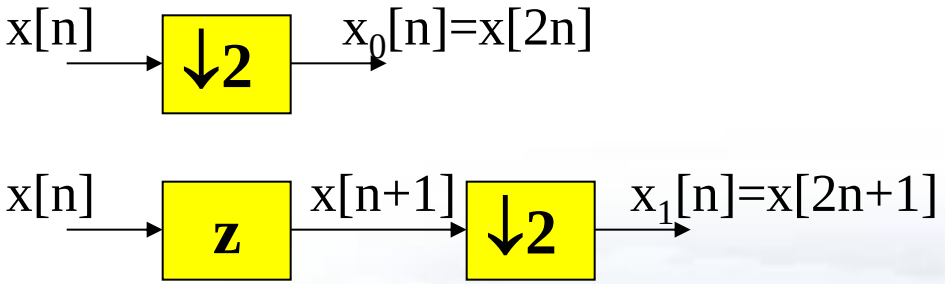
$$X_1[\frac{N}{2}+k]=X_1[k]$$

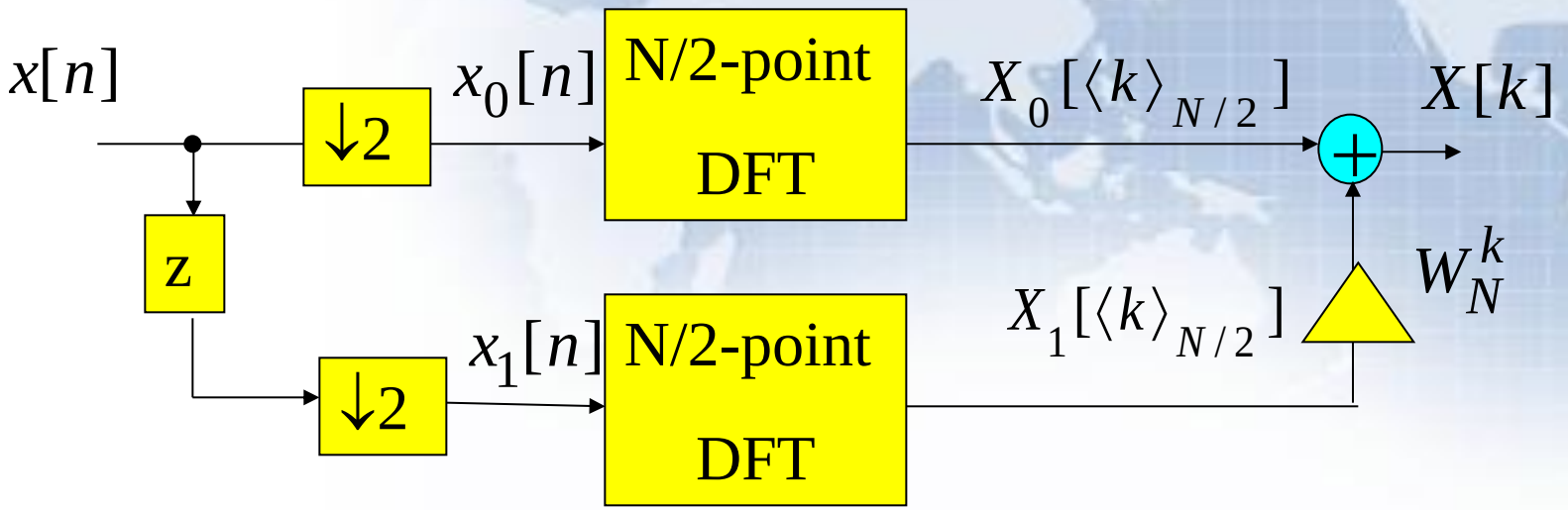
And: $$W_N^{(\frac{N}{2}+k)}=W_N^{\frac{N}{2}}W^k=-W^k$$

So:

$$X[\frac{N}{2}+k]=X_0[\frac{N}{2}+k]+W_N^{\frac{N}{2}+k}X_1[\frac{N}{2}+k]=X_0[k]-W_N^kX_1[k],\quad k=0,1,2,\cdots,N/2-1$$

- **Block-diagram interpretation:**





Decimation-in-Time FFT

- **The computation of N-point DFT by two methods:**

DFT computation	Complex Addition	Complex Multiplication
Direct computation	$N^2-N \approx N^2$	N^2
DIT to two N/2-point DFT	$(N^2/2)+N$	$(N^2/2)+N$

Decimation-in-Time FFT

- **Continuing the process we can express $X_0[k]$ and $X_1[k]$ as a weighted combination of two ($N/4$)-point DFTs.**

$$X_0[k] = X_{00}[\langle k \rangle_{N/4}] + W_{N/2}^{k} X_{01}[\langle k \rangle_{N/4}],$$
$$0 \le k \le (N/2) - 1$$

where $X_{00}[k]$ and $X_{01}[k]$ are the ($N/4$)-point DFTs of the ($N/4$)-length sequences:

$$x_{00}[n] = x_0[2n] \text{ and } x_{01}[n] = x_0[2n+1]$$

Decimation-in-Time FFT

- **Likewise, we can express**

$$X_1[k] = X_{10}[\langle k \rangle_{N/4}] + W_{N/2}^{k} X_{11}[\langle k \rangle_{N/4}],$$

$$0 \le k \le (N/2) - 1$$

where $X_{10}[k]$ and $X_{11}[k]$ are the (N/4)-point DFTs of the (N/4)-length sequences:

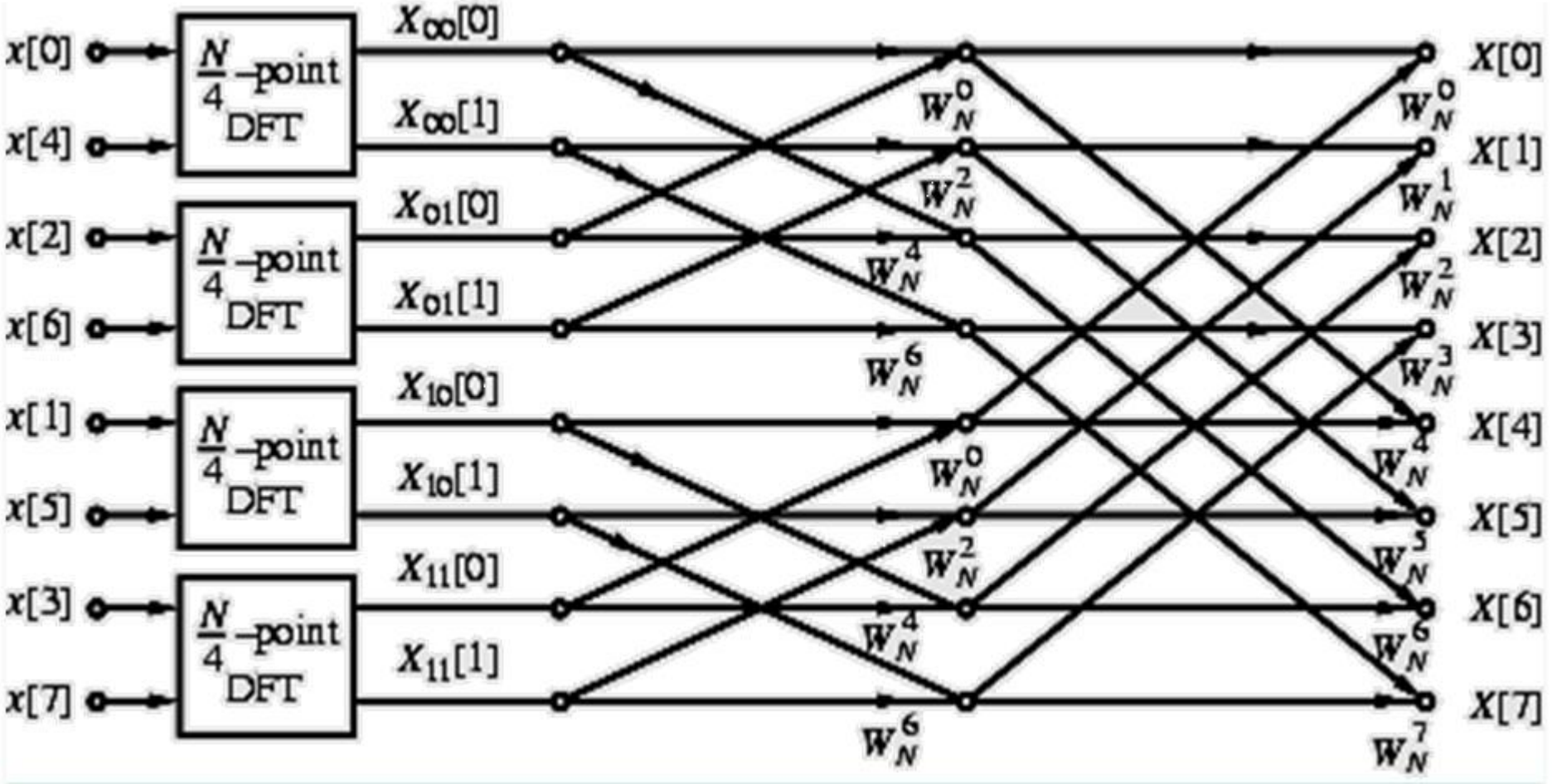
$$x_{10}[n] = x_1[2n] \text{ and } x_{11}[n] = x_1[2n+1]$$

Decimation-in-Time FFT

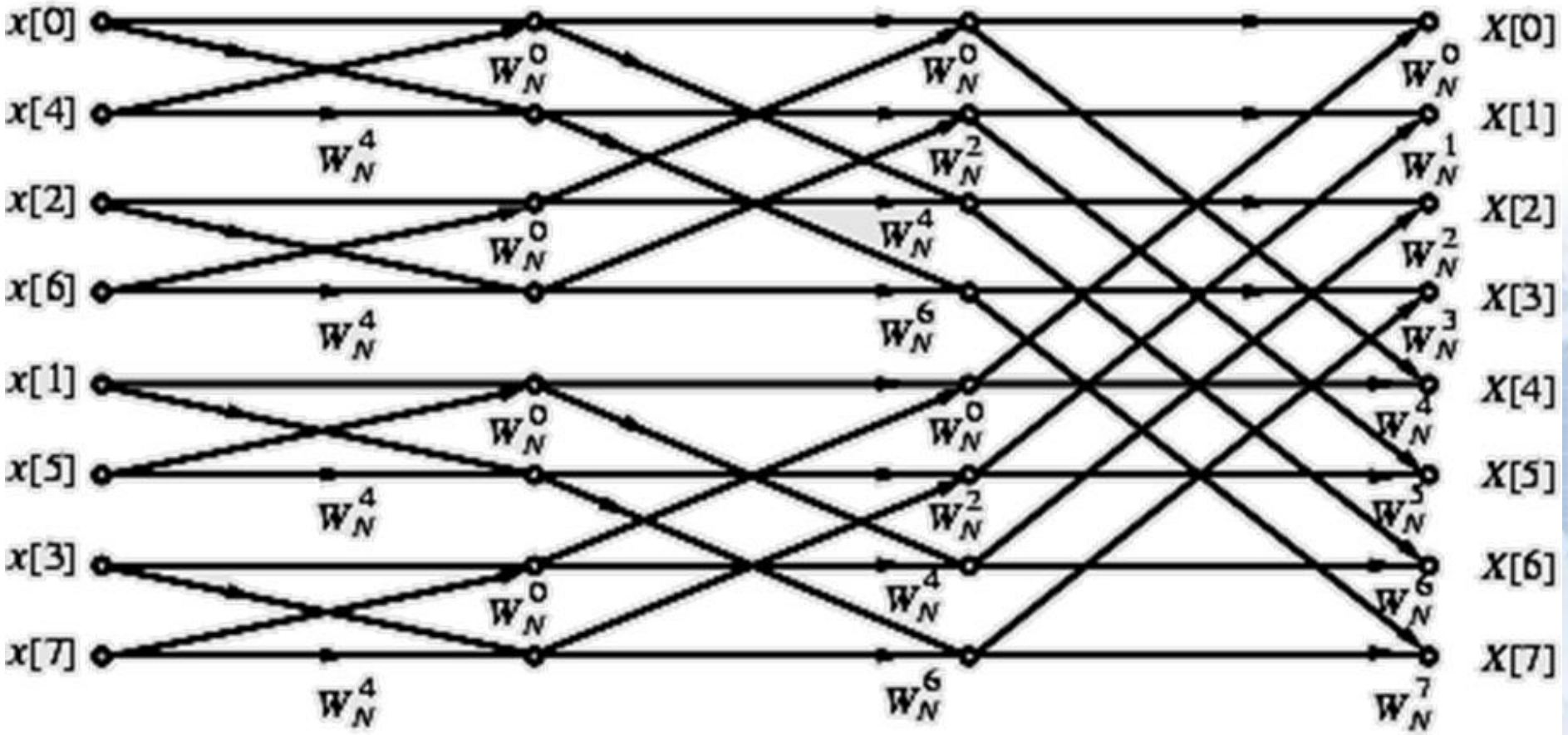
- **The following figure shows the time domain decomposition used in the FFT:**

Decimation-in-Time FFT

- **Flow-graph representation**



- **Complete flow-graph of the 8-point DFT is shown below**



Each stage of the DFT computation process employs the same basic computational module which is called a butterfly computation.

Decimation-in-Time FFT

- **Total number of complex multiplications and additions to compute all 8 DFT samples is equal to 8 + 8 + 8 = 24 = 8×3.**
- **In the general case when $N=2^k$, number of stages for the computation of the (2^k)-point DFT in the fast algorithm will be $k=\log_2 N$.**
- **Total number of complex multiplications to compute all N DFT samples is $N(\log_2 N)/2$**
- **Total number of complex additions to compute all N DFT samples is $N(\log_2 N)$**

Decimation-in-Time FFT

Number of Points	Direct Computation of the DFT		Radix-2 FFT	
	Complex Multiplies	Complex Additions	Complex Multiplies	Complex Additions
N	N^2	N^2-N	$(N/2)\log_2 N$	$N\log_2 N$
4	16	12	4	8
16	256	240	32	64
64	4096	4032	192	384
256	65536	65280	1024	2048
1024	1048576	1047552	5120	10240

Decimation-in-Time FFT

- **In developing the count, multiplications with $W_N^0 = 1$ and $W_N^{N/2} = -1$ have been assumed to be complex**

- **We have not taken advantage of the symmetry property:**

$$W_N^{(N/2)+k} = -W_N^k$$

These properties can be exploited to reduce the computational complexity further.

Decimation-in-Time FFT

- **The FFT algorithm described here also is efficient with regard to memory requirements;**
- **At the end of computation at any stage, output variables can be stored in the same registers previously occupied by the corresponding input variables;**
- **This type of memory location sharing is called in-place computation resulting in significant savings in overall memory requirements.**

Decimation-in-Time FFT

- In the DFT computation scheme outlined, the DFT samples $X[k]$ appear at the output in a sequential order while the input samples $x[n]$ appear in a different order;
- Thus, a sequentially ordered input $x[n]$ must be reordered appropriately before the fast algorithm described by this structure can be implemented.
- bit-reversed order

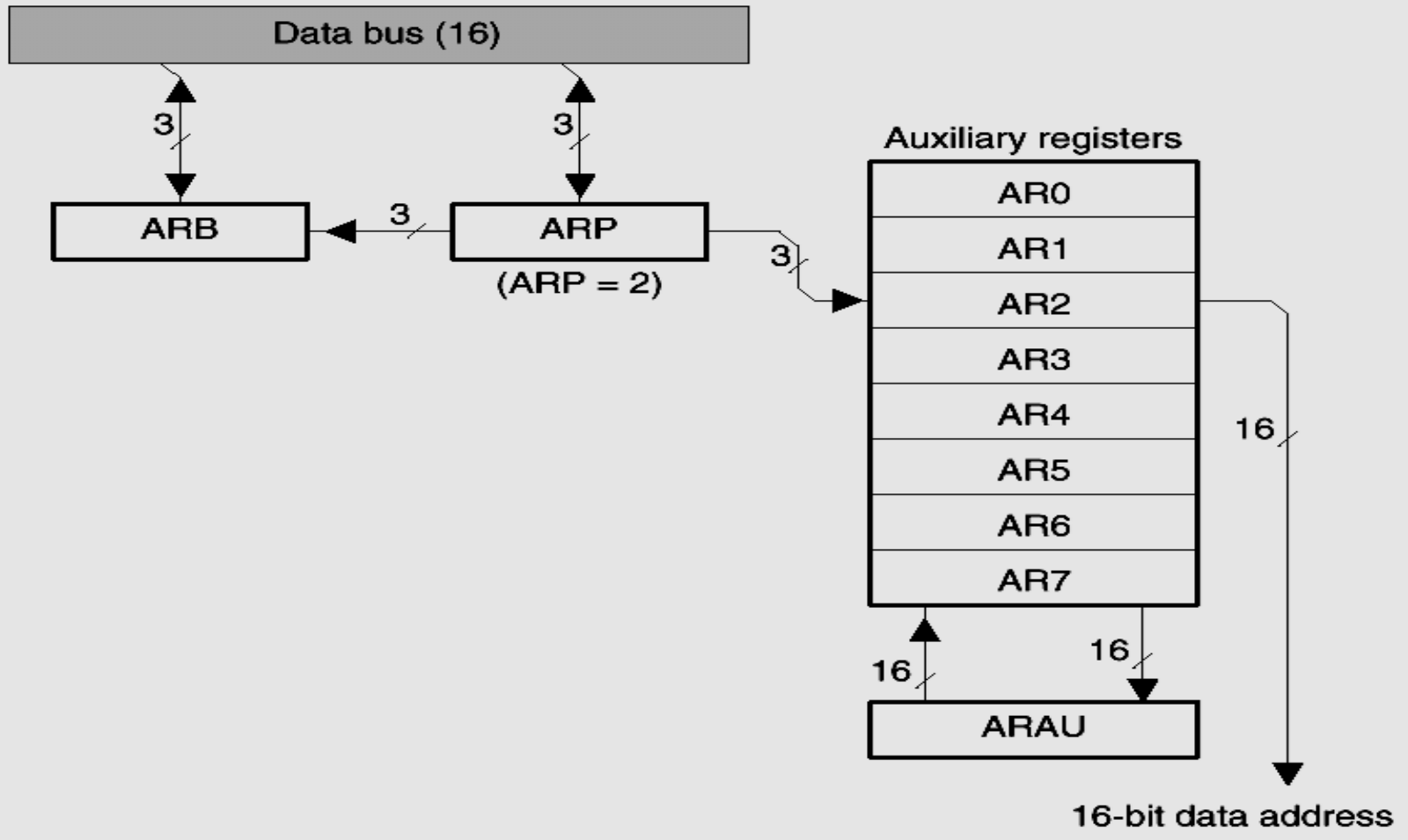
  if n=($b_2b_1b_0$) represents the index of $x[n]$, then m=($b_0b_1b_2$) is the correct index for FFT input samples. i.e.

  n: 000 001 010 011 100 101 110 111

  m: 000 100 010 110 001 101 011 111

original order	original address	bit-reversed address	bit-reversed order
0	000	000	0
1	001	100	4
2	010	010	2
3	011	110	6
4	100	001	1
5	101	101	5
6	110	011	3
7	111	111	7

Indirect Addressing in DSP



- **Bit-reversed addressing enhances execution speed and program memory for FFT.**

Decimation-in-frequency FFT

- **Given a sequence *x[n]* whose length is $N=2^L$, L is an integer. And dividing the sequence into two half sequences and compute its DFT:**

$$X[k]=DFT\{x[n]\}=\sum_{n=0}^{N-1}x[n]W_N^{nk}=\sum_{n=0}^{N/2-1}x[n]W_N^{nk}+\sum_{n=N/2}^{N-1}x[n]W_N^{nk}$$

$$=\sum_{n=0}^{N/2-1}x[n]W_N^{nk}+\sum_{n=0}^{N/2-1}x[n+\frac{N}{2}]W_N^{(n+\frac{N}{2})k}$$

$$=\sum_{n=0}^{N/2-1}[x[n]+x[n+\frac{N}{2}]W_N^{\frac{N}{2}k}]\bullet W_N^{nk},k=0,1,...,N-1$$

Decimation-in-frequency FFT

Because $W_N^{\frac{N}{2}} = -1$ **, and** $W_N^{\frac{N}{2}k} = (-1)^k$ **, then:**

$$X[k] = \sum_{n=0}^{N/2-1} [x[n] + (-1)^k x[n + \frac{N}{2}]] \bullet W_N^{nk}, k = 0,1,..., N-1$$

When k is even number:

$$X[2r] = \sum_{n=0}^{N/2-1} [x[n] + x[n + \frac{N}{2}]] \bullet W_N^{2rn}, k = 0,1,..., N-1$$

When k is odd number:

$$X[2r+1] = \sum_{n=0}^{N/2-1} [x[n] - x[n + \frac{N}{2}]] \bullet W_N^{(2r+1)n}, k = 0,1,..., N-1$$

Decimation-in-frequency FFT

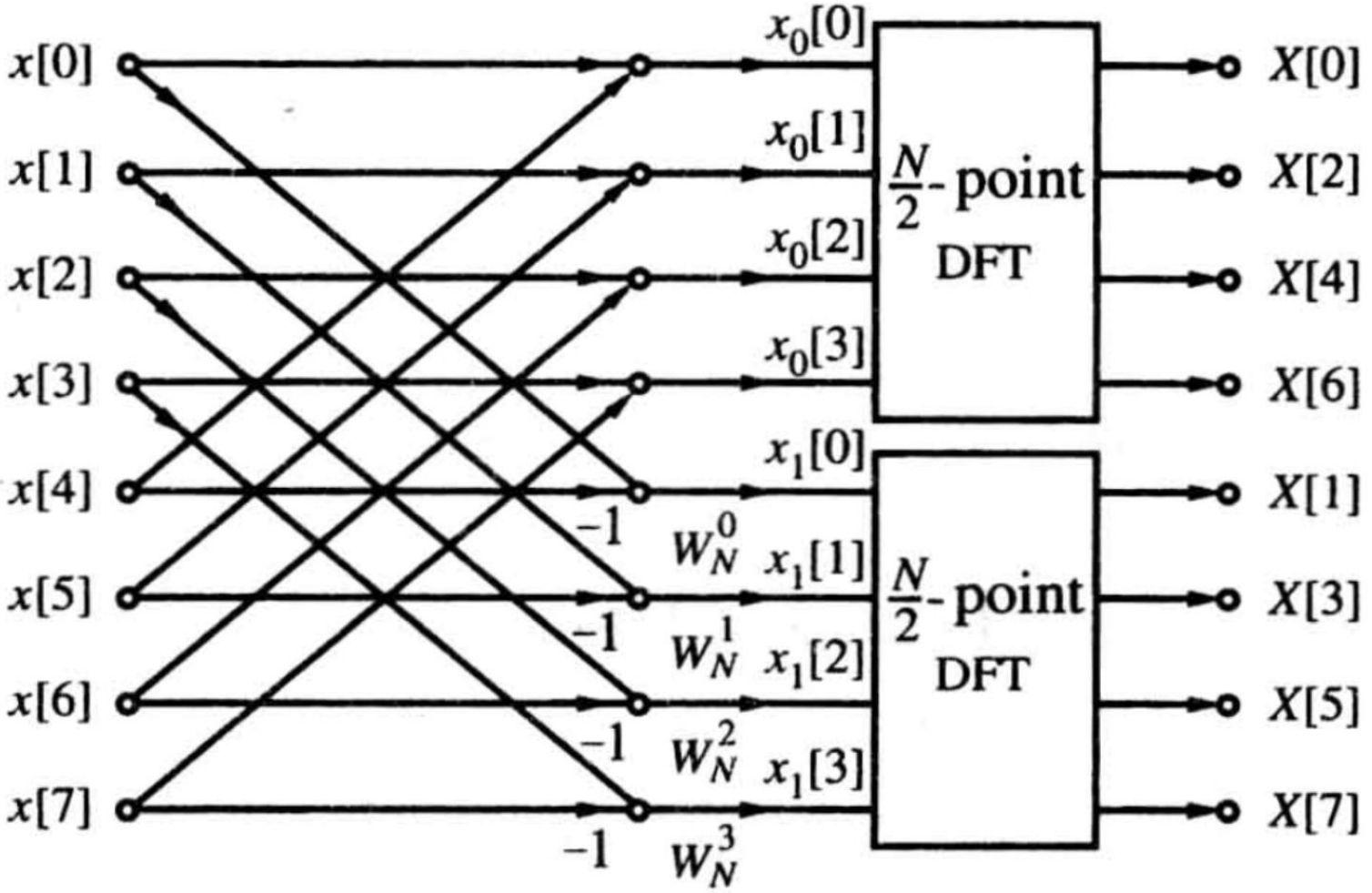
Based on $W_N^{2rk} = W_{N/2}^{rk}$, above formula equals to:

$$X[2r] = \sum_{n=0}^{N/2-1} [x[n] + x[n+\frac{N}{2}]] \bullet W_{N/2}^{rn}, r = 0,1,..., N/2-1$$

$$X[2r+1] = \sum_{n=0}^{N/2-1} \{[x[n] - x[n+\frac{N}{2}]] \bullet W_N^n\} W_{N/2}^{nr}, r = 0,1,..., N/2-1$$

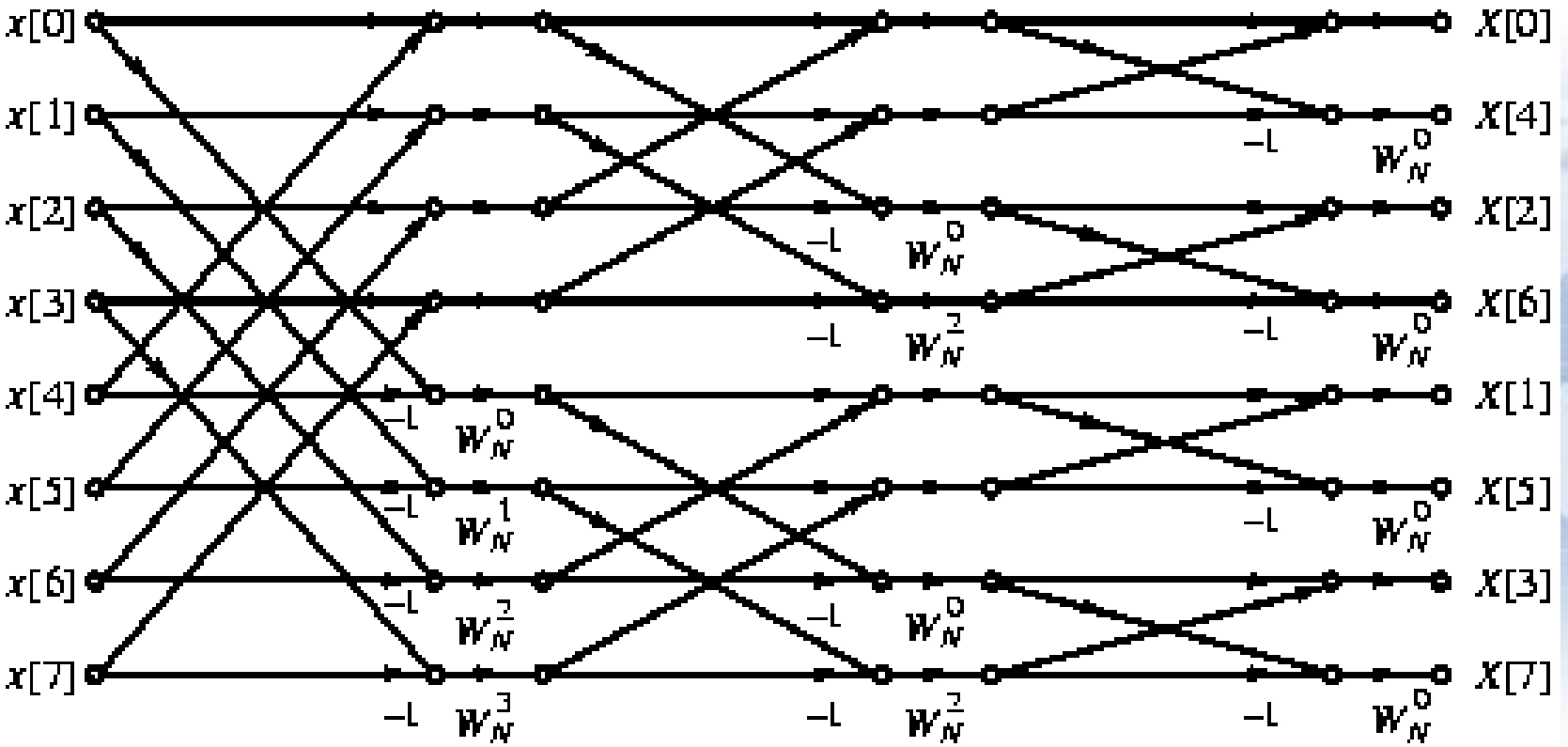
- **Process is continued until the smallest DFTs are 2-point DFTs.**

Decimation-in-frequency FFT



Decimation-in-frequency FFT

- **Complete flow-graph of the DIF FFT computation scheme for $N = 8$.**



Decimation-in-frequency FFT

- **Computational complexity of the radix-2 DIF FFT algorithm is same as that of the DIT FFT algorithm.**
- **Various forms of DIF FFT algorithm can similarly be developed.**
- **It can be easily seen that the flow-graph of DIF FFT is the transpose of the flow-graph of DIT FFT, and vice-versa.**

Inverse DFT Computation

- **An FFT algorithm for computing the DFT samples can also be used to calculate the inverse DFT (IDFT) efficiently.**
- **Consider a length-*N* sequence *x*[*n*] with an *N*-point DFT *X*[*k*]:**

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-nk}$$

Inverse DFT Computation

- **Multiplying both sides by *N* and taking the complex conjugate we get:**

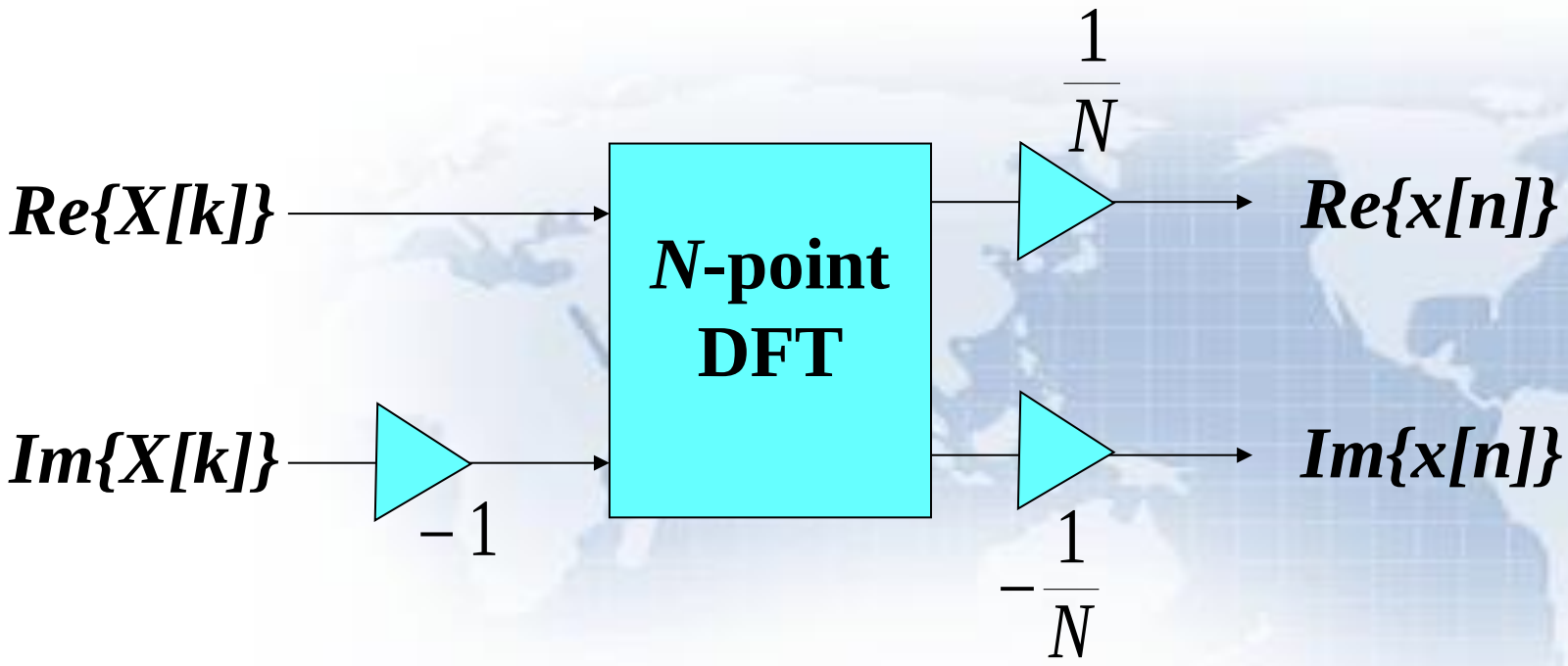
$$Nx^*[n] = \sum_{k=0}^{N-1} X^*[k] W_N^{nk}$$

- **Right-hand side of above is the N-point DFT of X*[k].**
- **Desired IDFT *x*[*n*] is then obtained as:**

$$x[n] = \frac{1}{N} \left\{ \sum_{k=0}^{N-1} X^*[k] W_N^{nk} \right\}^*$$

Inverse DFT Computation

Inverse DFT computation is shown below:



11.5 DFT and IDFT Computation Using MATLAB

- **Program 11_10.m**
  - Spectral Analysis
- **Program 11_11.m**
  - Linear Convolution

11.8 Number Representation